

● MEDIUM

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

02

10 questions · ~15 min

Q1

ADVANCED MATH

$$v^2 = \frac{LT}{m}$$

The formula above expresses the square of the speed v of a wave moving along a string in terms of tension T , mass m , and length L of the string. What is T in terms of m , v , and L ?

A. $T = \frac{mv^2}{L}$

B. $T = \frac{m}{v^2L}$

C. $T = \frac{mL}{v^2}$

D. $T = \frac{L}{mv^2}$

Q2

GRID-IN

ADVANCED MATH

$$y = 4x$$

$$y = x^2 - 12$$

A solution to the given system of equations is x, y , where $x > 0$. What is the value of x ?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x^2 + y + 10 = 10$$

$$8x + 16 - y = 0$$

The solution to the given system of equations is x, y . What is the value of x ?

- A. -16
- B. -4
- C. 2
- D. 8

$$x + y = 17$$

$$xy = 72$$

If one solution to the system of equations above is (x, y) , what is one possible value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which quadratic equation has exactly one distinct real solution?

A. $x + 15^2 = 0$

B. $x + 15^2 = -45$

C. $x + 15^2 = 45$

D. $x + 15^2 = 135$

Q6

GRID-IN

ADVANCED MATH

$$38x^2 = 389$$

What is the negative solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GRID-IN

ADVANCED MATH

$$d - 30d + 30 - 7 = -7$$

What is a solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$P = N19 - C$$

The given equation relates the positive numbers P , N , and C . Which equation correctly expresses C in terms of P and N ?

A. $C = \frac{19 + P}{N}$

B. $C = \frac{19 - P}{N}$

C. $C = 19 + \frac{P}{N}$

D. $C = 19 - \frac{P}{N}$

Q9

GRID-IN ADVANCED MATH

$$x^2 - 5x - 24 = 0$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$v = -\frac{w}{150x}$$

The given equation relates the distinct positive numbers v , w , and x . Which equation correctly expresses w in terms of v and x ?

- A. $w = -150vx$
- B. $w = -\frac{150v}{x}$
- C. $w = -\frac{x}{150v}$
- D. $w = v + 150x$